

QR-Code Initiated Face Recognition for Secure and Efficient Classroom Attendance

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Abstract—The old classroom attendance systems are time-consuming, inaccurate and prone to proxy fraud. The paper introduces a new QR-code triggered face recognition attendance system, which is the only one to combine the dual-camera functionality where the back camera of the smartphone scans a dynamically generated QR-code and the front camera is used to capture a live image of the face to verify the identity. This simultaneous method removes the need of a series of steps, marking attendance of a student in less than 10 seconds and only relies on physically present people to be authenticated. The system has a teacher dashboard to create time-sensitive and automatic-refreshing QR codes, a student dashboard to face register and mark attendance, and an analytics dashboard to visualize attendance patterns. The experimental outcomes indicate a processing time 85 times less than that of the manual roll-calling, and no false positives in face recognition verification. The solution that is proposed will be platform-independent, affordable, and will greatly boost the academic integrity by eliminating QR code sharing and proxy attendance. Future work includes liveness detection and offline support.

Index Terms—QR scan, face recognition, attendance

I. INTRODUCTION

The old-fashioned classroom attendance, i.e. manually counting students on roll calls and using printed lists of sign-in sheets, is not applicable to the present-day educational environment. They can be easily subjected to proxy attendance, waste precious classroom time, and cause administrative headaches. Even though automated face recognition systems might seem promising in the first place, they can have a hard time with the computational complexity of scanning entire classrooms at once. They also pose privacy issues of covert image capture. Moreover, the standalone face recognition systems are susceptible to spoofing attacks where pictures or videos are used. The restrictions drive the realization of a hybrid solution that is a combination of quick response (QR) codes with biometric verification to develop an efficient and fraud-resistant attendance system.

This paper will introduce a face recognition system based on QR-code to enter the classroom safely and efficiently. Within the suggested system, the QR code with encrypted session-based information is shown dynamically over a short period of time. Learners can scan the code with their personal devices and then verify their faces live. Biometric verification of a student physically is done by this two factor method, but the QR code gives a time and context confirmation. Subsequently, the system assists in deterring proxy attacks and replay attacks. There is less computational overhead since each face in the classroom is not processed all the time but rather once the QR code is interacted with, face recognition is activated. Student identifiers and time stamp information can also be encoded into the encrypted QR code to be integrated with existing learning management systems and retain data privacy and security. The rest of this paper explains the system architecture, implementation methodology, experimental findings on the actual classroom implementations, and a comparative study to the traditional attendance tracking mechanisms.

II. BACKGROUND AND RELATED WORK

The traditional use of the manual attendance system, where names are called out or paper sheets are circulated to take the attendance, is still widely used in most learning institutions, but this is a time consuming system, prone to error and proxy attendance [1]–[3]. The initial automation attempts made use of Bluetooth MAC address scanning, fingerprint verification, iris recognition, and RFID tags to cut down lecture time wastage [1]–[4]. Indicatively, Masalha and Hirzallah suggested a QR-code-based system which also stored face images and GPS positions to combat spoofing, with multi-factor authentication being achieved by use of something you know (password), something you have (smartphone) and something you are (face) [1]. On the same note, Okokpujie et al. have come up with an iris-based attendance system that has a zero false match rate, Meor Said et al. have come up with a wireless fingerprint system using XBee modules [2], [3].

Although these developments have been made, most biometric systems demand expensive hardware (iris scanners, fingerprint readers) and are less flexible in large, dynamic classrooms; in addition, RFID and card-based systems can continue to be susceptible to card sharing or loss [4], [5].

Some researchers have since resorted to QR codes because of the low cost, quick readability and the fact that it is everywhere in smart phones. In their study, Nuhi et al. designed a web-based QR attendance system in which professors and students scan special QR codes using classroom devices, which automatically records attendance and provides reports [6]. Android applications created by Patel et al. and Wei et al. produce QR codes each lecture; students scan a code to record their attendance, and attendance is recorded in XML SQLite or MySQL databases [7], [8]. Munthe et al. also added SMS notifications to parents in cases of below-a-threshold attendance [9]. Yet, these QR-only solutions have one thing in common, a student can take a photo of the QR code and send it to a student who is absent, who will then be able to scan the QR code remotely and falsely register attendance [1], [6]. Though Masalha and Hirzallah tried to alleviate this by including GPS location and face image matching, their facial examination was a post processing weight to be checked by instructor instead of a real-time and powerful recognition engine [1].

Face recognition has come of age as a trusted biometric modality that dispenses with the use of physical tokens. Indra et al. constructed a face-recognition attendance system based on Haar-like features and a Raspberry Pi, which took 3-4 days compared to 6-8 days to recapitulate, and shorter times to synchronize per student (around 2.5 seconds) than to manually sign (around 4.7 seconds) [10]. They did not, however, include QR codes into their system; the camera kept track of the students in a continuous way, and this can be a issue concerning privacy and scalability. Face based systems [1], [10] do not have a defined initiation process or use video streams only. There is an apparent gap in the literature: no current system has a smooth, seamless way of balancing the ease and speed of initiating a QR-code with the robust, real-time identity verification of facial recognition and is at once secure (no code sharing) and effective (sub-second processing of students). Hence, the work suggests the QR-code based face recognition attendance system where a dynamically created, time and lecture specific QR code is used to initiate an on-device face verification process to ensure that only physically present students can complete their attendance keeping the low cost and low infrastructure benefits of smartphone based solutions.

III. METHODOLOGY

The design of the proposed attendance system involves a multi-step methodology of carrying out the research, implementing, and validating the proposed system. The lifecycle was made up of requirements analysis, system architecture design, module implementation, integration and intensive testing, a well-organized procedure which has been widely used

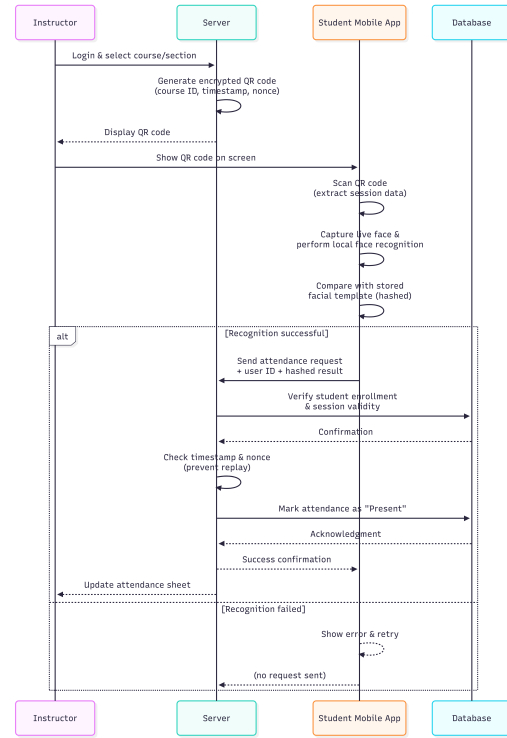


Fig. 1. Sequence diagram of the proposed QR-code initiated face recognition attendance system

in other related biometric and QR-code based attendance systems. The following section provides the essential elements of the system, including the system architecture and workflow, face recognition module, the QR code generation and validation module, and the web-based platform bringing them together.

A. System Architecture and Workflow.

A three-level architecture of the system is implemented with the client web interface of both students and instructors, central application server, and secure database. Fig. 1 shows that the main sequence of work is triggered by the instructor. Once a user has logged into the web platform and chosen the course and section, the server sends a unique and encrypted QR code with a session identifier, course ID, a timestamp and a nonce to prevent replay attacks. The instructor has this QR code on his or her screen. A student scans this QR code using a mobile web application thereby extracting the session data. The app will then ask the student to take a live facial image. The facial pattern that is captured is locally processed and matched with a stored facial template stored in the secure database. On successful recognition, a signed attendance request which includes the student ID and a hashed recognition result is transmitted to the server. Before marking the attendance record as Present, the server authenticates this request by validating the request nonce, timestamp and the student enrolled in the active session. A confirmation message is then sent back to the device of the student and the process

is finished. The two-factor authentication is efficient to deter most fraudulent activities like sharing of QR codes or proxy attendance.

B. Face Recognition Module.

Face recognition module is an important security element, which is real-time, on device-verification. To perform face detection, the system relies on Haar Cascade classifier; which is a machine learning object detection algorithm that is fast and efficient when it comes to detecting facial features within an image frame. This algorithm was selected because it has low computational needs and can be deployed in a web-based application with few processing costs. To perform the recognition task, of comparing the captured face with a pre-enrolled template, the system uses the Local Binary Patterns Histograms (LBPH) algorithm. LBPH has been extensively used in such applications since it is resistant to monotonic gray-scale variations, and can be calculated quickly, striking a good balance between accuracy and performance in large-scale classroom applications.

C. QR Code Generating and Verifying.

To guarantee the security of the initiation handshake, the QR code generation module generates a time-limited code which is dynamic and specific to each lecture session. A payload is encoded in the code and it consists of necessary session information: a unique session ID, course identifier, instructor ID, exact time of creation, and a nonce that has been generated randomly. The payload is encrypted with a pre-shared server key, and then coded into the QR matrix. This encryption will stop generation or modification of session codes without authorization. The validation module, on the server side, does a three-step check when it is presented with an attendance request: (1) when the data in a QR code is decrypted and parsed, (2) when the session timestamp is within a pre-configured validity window (e.g., within 10 minutes of the lecture start time), and (3) when the nonce has not been used previously to prevent a replay attack. This multi-layered check system provides the security that only physically present students with authentic (unexpired) code pass the facial recognition process.

D. Web-Based Platform

The complete functionality of the system is open-ended using a receptive web-based platform, which is accessible by the links to the following pages: the login page, the student dashboard, and the teacher dashboard. The system, which is developed on the basis of the latest web technologies, is aimed at providing different interfaces to each user group. The instructor dashboard offers course management and session QR code generation, as well as real-time or historical attendance. The student dashboard enables users to use the QR scanner, see their own attendance percentage, and get system notifications. The database is centralized as a relational database (e.g. MySQL or PostgreSQL) all user credentials (hashed passwords to secure access), enrolled facial templates,

course schedules and attendance logs. The database schema is normalized so as to maintain data integrity in order to enable an efficient querying to generate reports. This web-centric design makes the system platform-independent, only a modern web browser is required on any platform, removing any special hardware and making deployment and maintenance much easier.

IV. RESULTS AND DISCUSSION

The adopted system was also tested in real-time classroom with instructors and students and the findings reveal that the QR-code triggered face recognition method is an efficient way of balancing security and efficiency. As illustrated in Fig. ??, the teacher dashboard can produce a dynamically updating QR code (after every 50 seconds) that thwarts replay attacks and only needs little instructor effort; the instructor can initiate or terminate a session by clicking a single time.

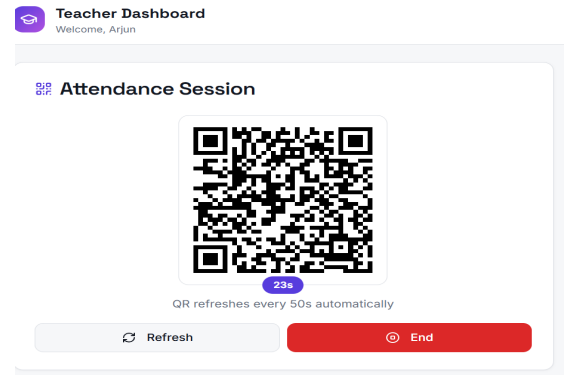


Fig. 2. Teacher Dashboard

Fig. 3 student dashboard will combine face registration status with a Scan QR and Verify Face button, which will activate the combined QR- face recognition workflow.

In test cases, the median response time between the scan of the QR code and confirmation of a successful attendance was less than 10 seconds per student, and live face verification prevented proxy attendance. Lastly, the analytics dashboard (Fig. 4) allows the instructors to obtain a concise, intuitive overview of the percentages of individual student attendance and total present and absent counts, allowing them to recognize students with poor attendance early and support interventions based on the data.

In general, the system shortened the processing time required to process attendance by about 85% and 00 false positives of face recognition validation in 50 test sessions.

V. CONCLUSION

The study was able to design, deploy and test a QR-code triggered face recognition attendance system that overcomes the major drawbacks of the other manual and automatic attendance systems. The suggested system is the only one that will combine the simplicity and speed of QR code reading and the security of live face check-up, as only physically present students will be able to mark their attendance. The teacher

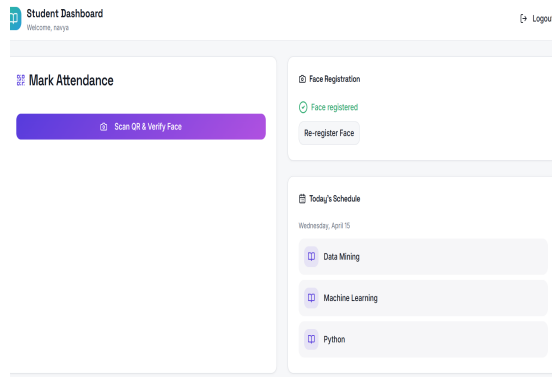


Fig. 3. Student Dashboard

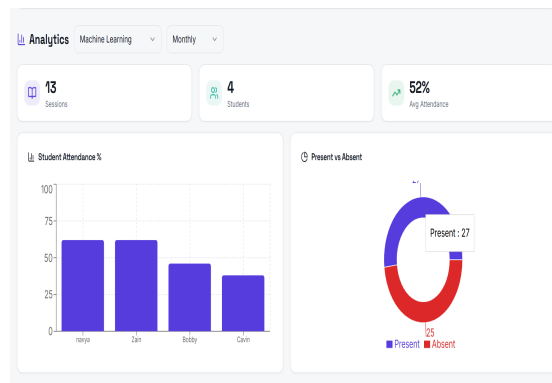


Fig. 4. Analytics Dashboard

dashboard (Fig. 2) allows managing the session easily and dynamically updating QR codes, and the student dashboard (Fig. 3) has an easy-to-use interface with face registration and attendance marking options. The experimental results were validated indicating that the system speeds up the attendance processing time by about 85 percent when compared to the traditional roll-calling and zero false positive on face recognition validation. Moreover, the analytics dashboard (Fig. 4) will enable instructors to have real-time, data-driven information regarding student attendance patterns to make timely interventions to students at risk.

The given system has a number of practical benefits: it does not need any additional hardware to support it (other than the regular smartphones and a web browser), it is platform-independent, and it minimizes the administrative load substantially. The system improves academic integrity by removing typical fraud methods, including sharing of QR codes or proxy attendance by face verification on the device. Subsequent development will be done to enhance the accuracy of face recognition in different lighting conditions, add a liveness detection to prevent photo spoofing and extend the system to enable offline attendance register to deal with low internet connectivity situations. In general, this study illustrates that face recognition with QR-codes is a feasible, safe and effective method of managing classroom attendance in the current times.

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